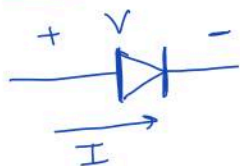


DIODES: (from chapter 2 of textbook)

1

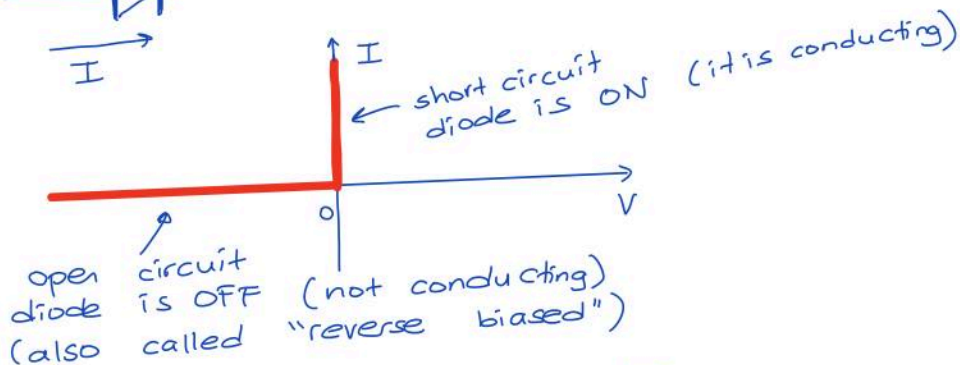
- nonlinear resistors
- resistance depends on the voltage across the diode

Ideal Diode:

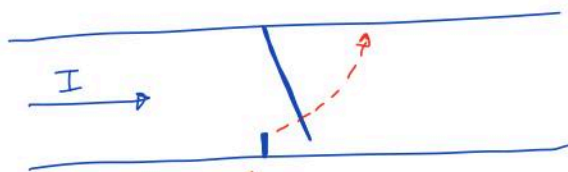


$$V=0, \text{ when } I > 0 \Rightarrow \text{short circuit}$$

$$I=0, \text{ when } V < 0 \Rightarrow \text{open circuit}$$



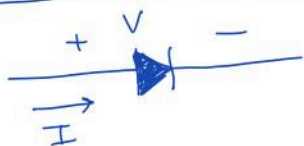
Water analogy: check valve



when flow is in the correct direction, valve opens fully

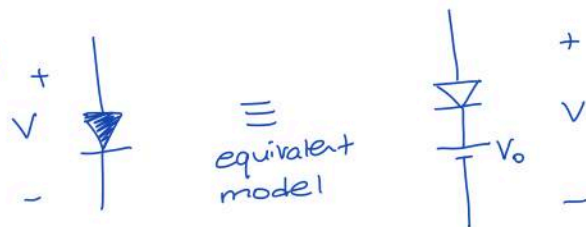
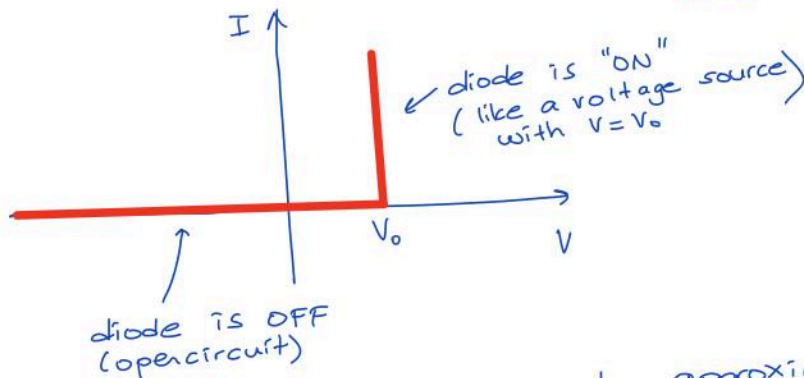
when flow is in reverse direction, does not let it flow.

A more realistic model for diode:



$$\left\{ \begin{array}{ll} V = V_0 & \text{when } I > 0 \\ I = 0 & \text{when } V < V_0 \end{array} \right.$$

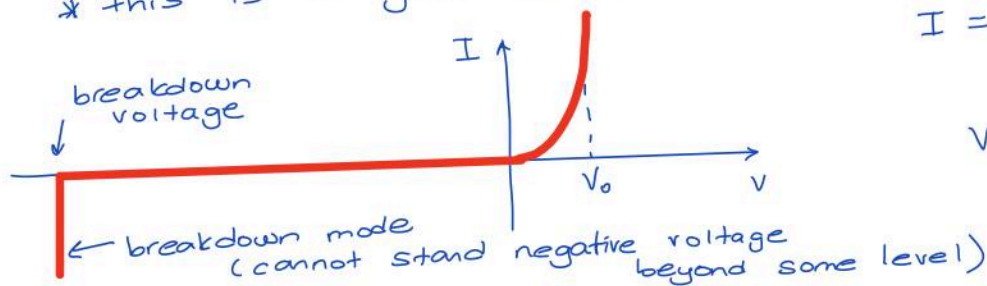
where V_0 : threshold voltage



* this is a good enough approximation for a real-life diode

$$I = I_s (e^{V/r} - 1)$$

$$V_0 \sim 0.7V$$



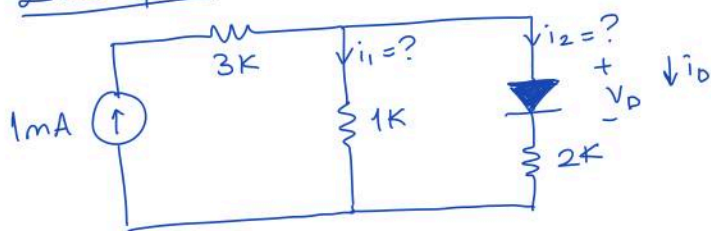
Solution of Circuits Containing Diodes

(2)

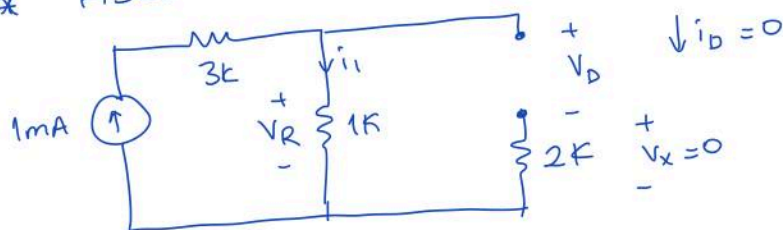
- 1) Assign ON or OFF state to the diode(s).
- 2) Replace diode(s) with the equivalent circuit in that state.
- 3) Solve the resulting linear circuit.
- 4) Check if the conditions of the assumed state is satisfied for the diodes.
- 5) If not, go to step 1 and change state of the diode(s).

Example:

Diode with $V_0 = 0.7V$



* Assume diode is OFF: diode $\Rightarrow$ open circuit



$$i_2 = 0$$

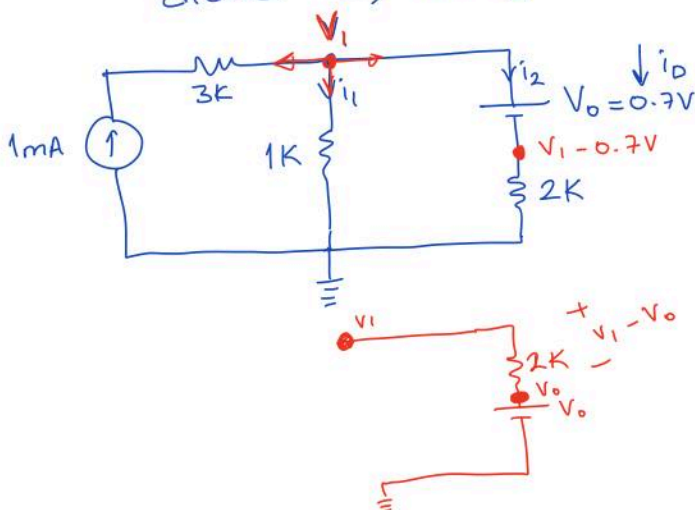
$$i_1 = 1mA$$

$$V_R = i_1 \times 1K = 1V$$

$$V_R = V_D + V_x = V_D = 1V > 0.7V$$

X cannot happen!
we need $V_D < 0.7V$ when diode is OFF.

* Assume diode is ON:
diode $\Rightarrow$ voltage source with $V = V_0$



$$\text{KCL: } -1mA + \frac{V_1}{1K} + \frac{V_1 - 0.7}{2K} = 0$$

$$3V_1 = 2.7V$$

$$V_1 = 0.9V$$

$$i_1 = \frac{V_1}{1K} = 0.9mA$$

$$i_2 = 1mA - i_1 = 0.1mA$$

$$= \frac{V_1 - 0.7}{2K} = 0.1mA$$

* check diode condition:

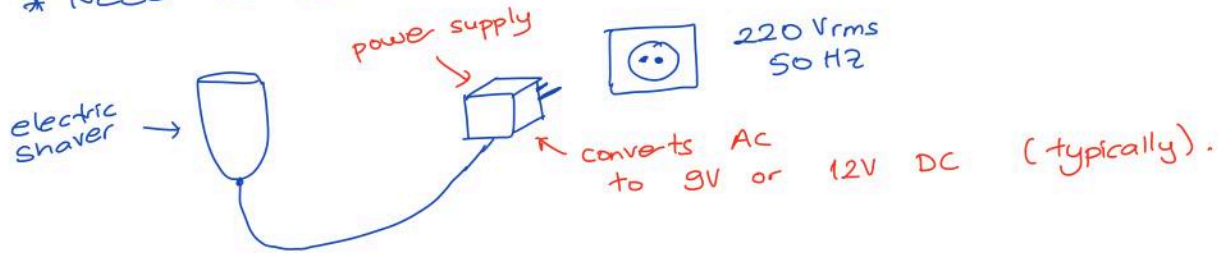
$$i_D = i_2 = 0.1mA > 0$$

condition satisfied.

Diodes as Rectifiers

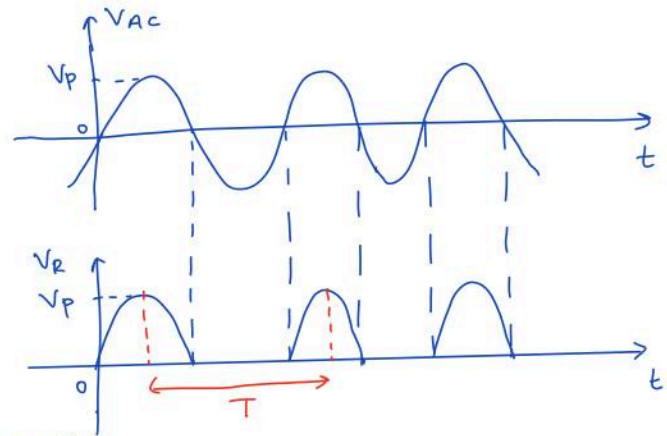
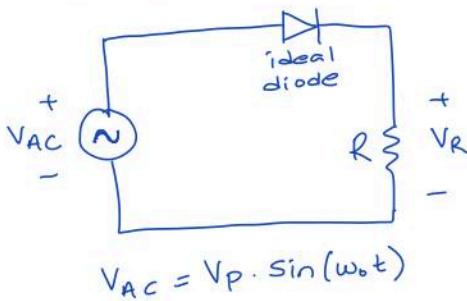
(3)

- * Diodes are used for "rectification"
- * Need to convert AC to DC for some appliances to work.

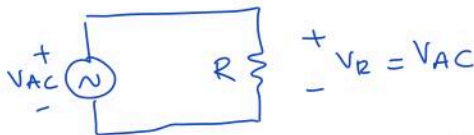


Converting AC to DC

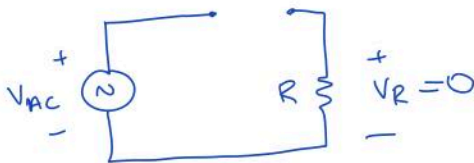
Half-wave Rectifier



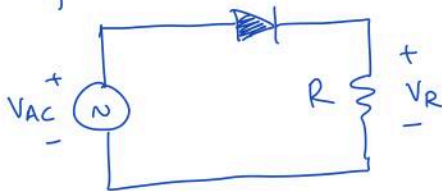
- * when $V_{AC} > 0 \Rightarrow$ diode turns ON
 $\Rightarrow$ acts like short circuit



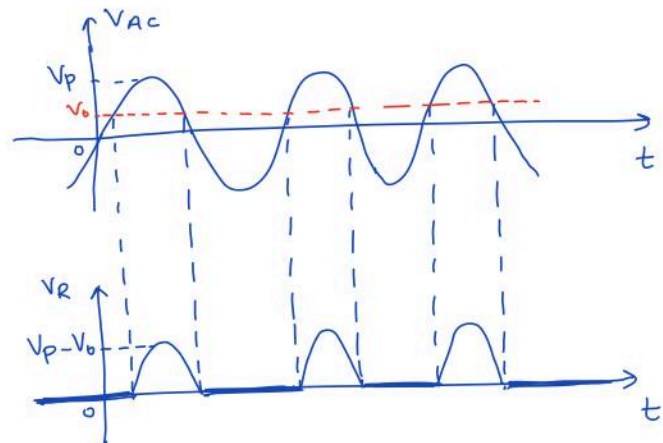
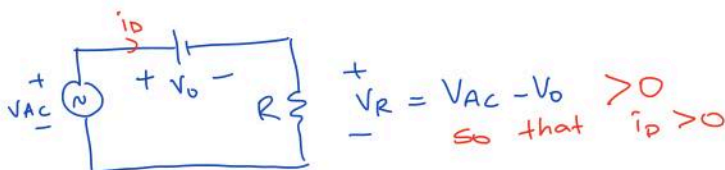
- * when $V_{AC} < 0 \Rightarrow$ diode turns OFF
 $\Rightarrow$ acts like open circuit



- * if we have a diode with non-zero V_0 :

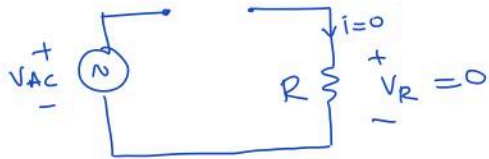


- * when $V_{AC} > V_0 \Rightarrow$ diode turns ON
 $V_D = V_0$

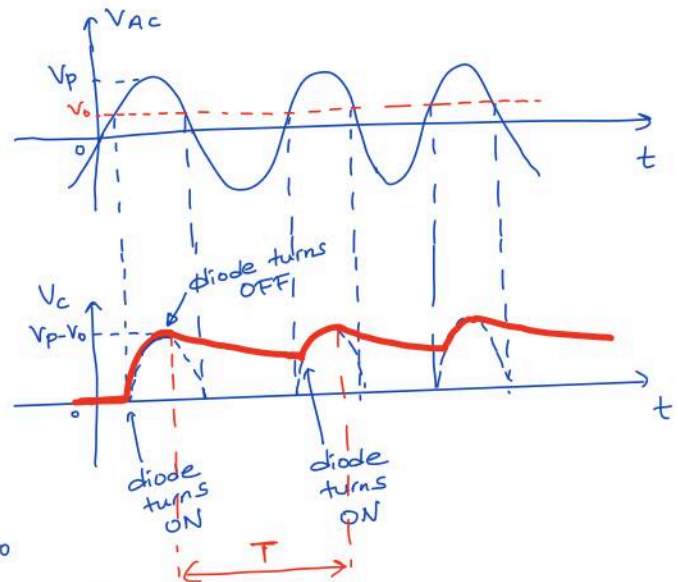
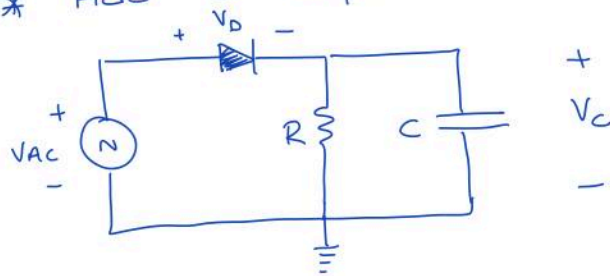


* when $V_{AC} < V_0 \Rightarrow$ diode turns OFF
 $\Rightarrow$ open circuit

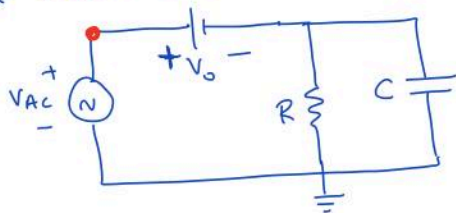
4



* Add a capacitor :



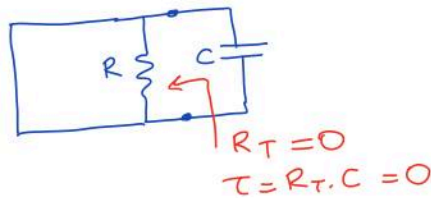
* when diode is ON :



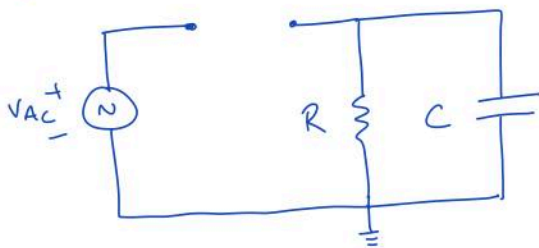
$$V_C = V_{AC} - V_0$$

$\Rightarrow$ capacitor follows input voltage and it charges

time constant:



* when diode is OFF :

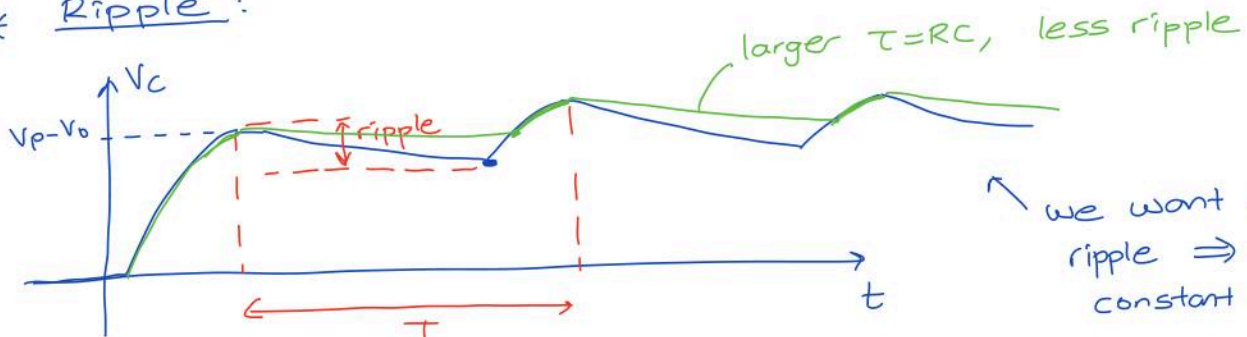


$$V_C = V_0 \cdot e^{-\frac{(t-t_0)}{\tau}} \text{ for } t > t_0$$

$\Rightarrow$ capacitor is discharging

$$\tau = RC$$

* Ripple :

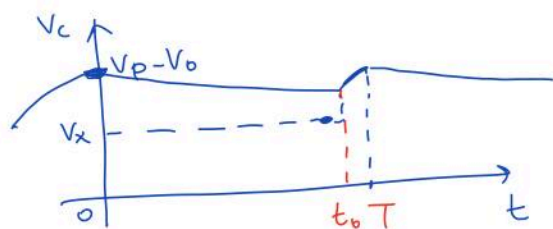


$\nwarrow$ we want small ripple $\Rightarrow$ almost constant voltage

water analogy:

rectifier → water tank in apartment buildings
 ripple → depends on the size of the tanks,
 & how much water we use,
 & duration of water interruption
 diode → prevents water leakage back to the city pipes

* Amplitude of ripple:



Assume $t_0 \approx T \Rightarrow$ very large $\tau = RC$.

$$V_x = (V_p - V_0)e^{-\frac{t_0}{\tau}} \approx (V_p - V_0)e^{-\frac{T}{\tau}}$$

* So, peak-to-peak ripple is:

$$V_{\text{ripple}} = (V_p - V_0) - V_x = (V_p - V_0) - (V_p - V_0)e^{-\frac{T}{\tau}}$$

$$= (V_p - V_0) \left[1 - e^{-\frac{T}{\tau}} \right]$$

Assume $\tau \gg T$
 $e^{-x} \approx 1 - x$ for small x

$$= (V_p - V_0) \left[1 - \left(1 - \frac{T}{\tau} \right) \right]$$

$$= (V_p - V_0) \cdot \frac{T}{\tau} = \boxed{(V_p - V_0) \cdot \frac{T}{RC}} \quad \text{if } \tau \gg T$$

* To reduce ripple:

big capacitance

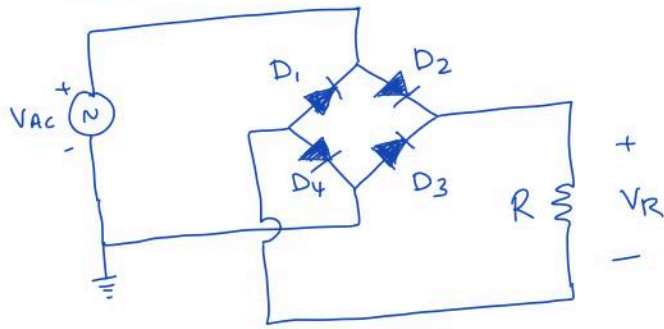
big resistance

short period $\Rightarrow$ cannot be controlled.

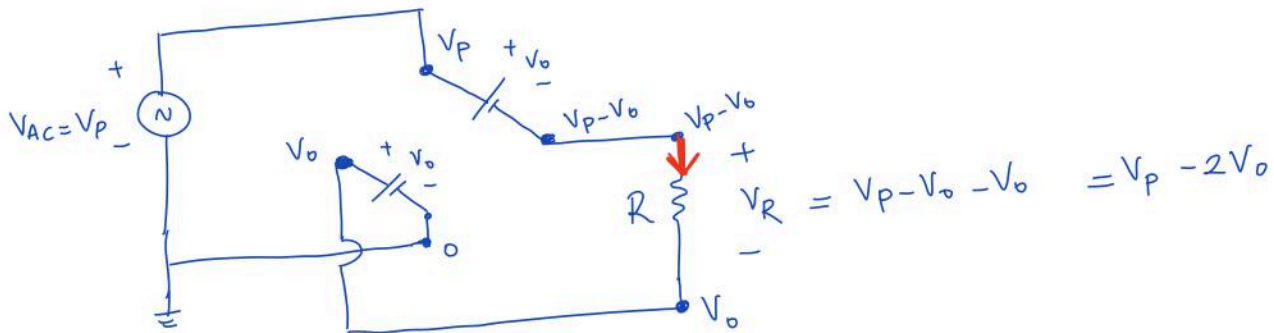
Full-wave Rectifier (Bridge Rectifier)

6

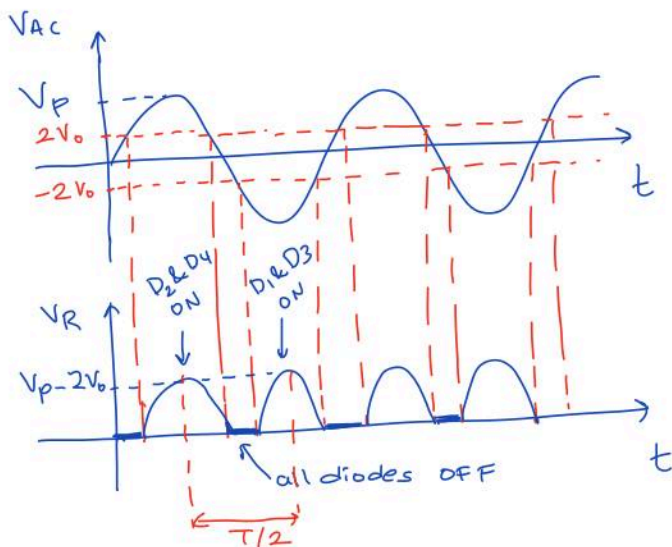
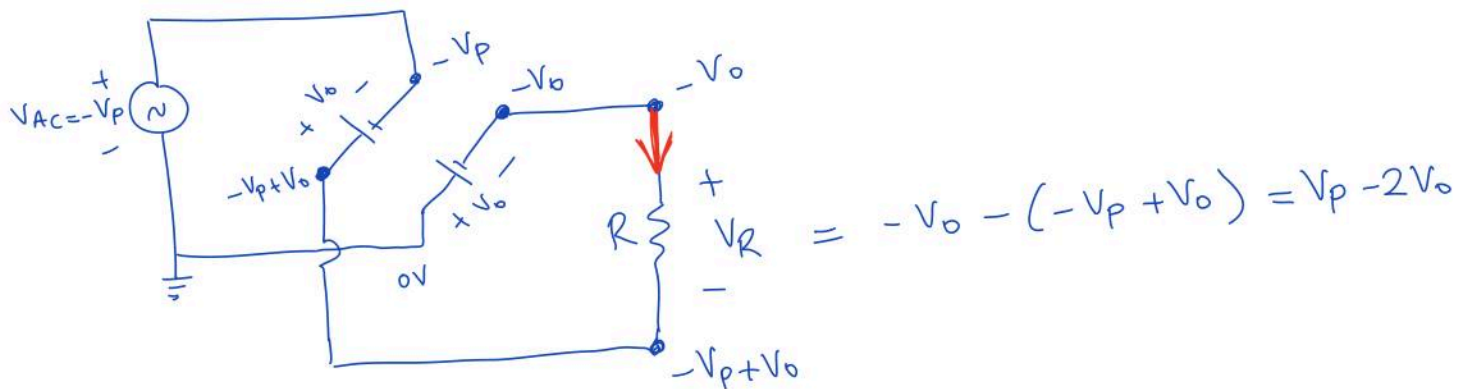
Uses both negative & positive cycles



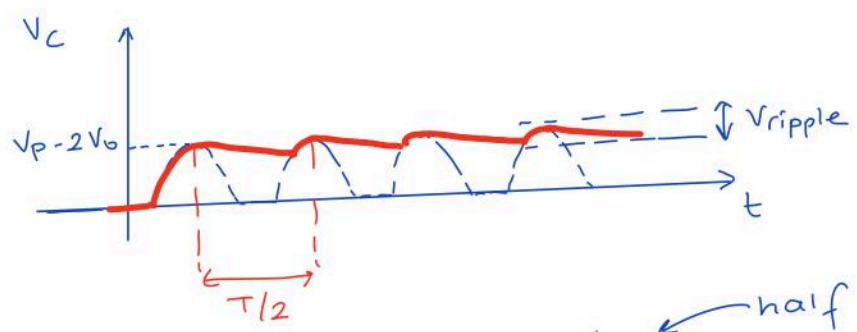
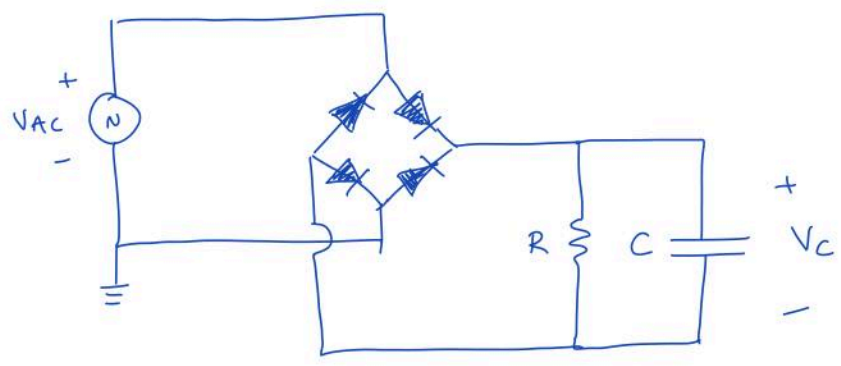
* When $V_{AC} = V_p$, D_2 and D_4 ON, D_1 and D_3 OFF



* When $V_{AC} = -V_p$, D_1 and D_3 ON, D_2 and D_4 OFF



* Now, add a capacitor

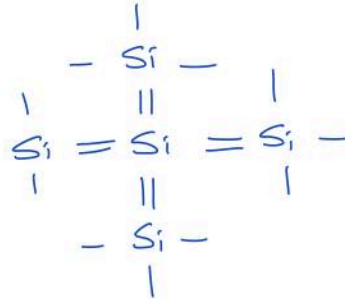


$$V_{ripple} = (V_p - 2V_0) \frac{T/2}{RC} \quad \leftarrow \text{half the period}, \text{ if } \tau \gg \frac{T}{2}$$

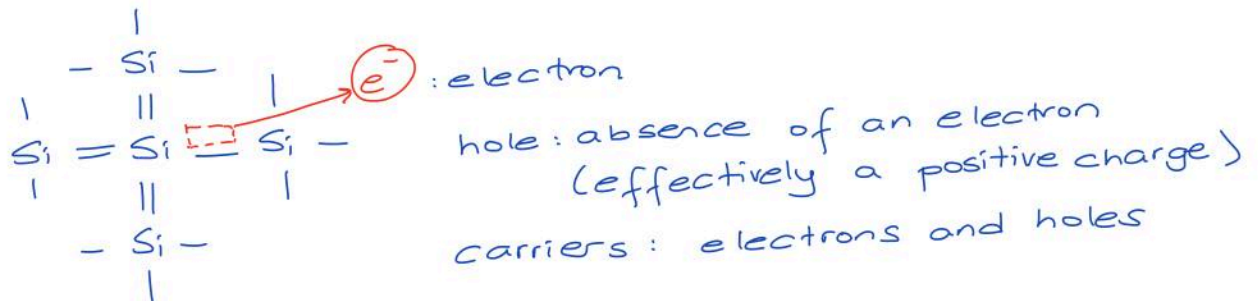
- * a better approach $\Rightarrow$ ripple reduces to half
- * uses 4 diodes instead of one

Silicon as a Semiconductor

- * Si : 4th column element $\Rightarrow$ 4 valence electrons
- * Si makes covalent bonds with 4 neighboring Si atoms



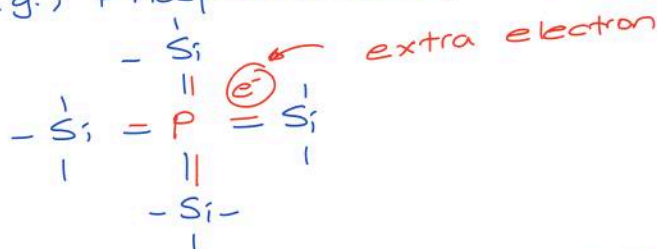
- * covalent bonds might be broken due to thermal energy



- * Another electron can jump into hole, causing hole position to change $\Rightarrow$ hole current
 $\Rightarrow$ analogy : moving air bubble
- * Pure crystalline silicon is a poor conductor

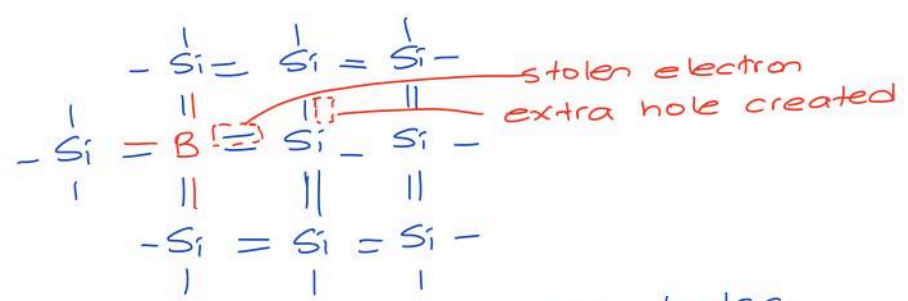
Doping: Replace Si atoms with dopants with extra e^- or hole, to increase conductivity

- * Donor $\Rightarrow$ n-type semiconductor
 e.g., Phosphorus (5A) : donates 1 extra electron



majority of carriers are these free electrons
 more doping $\Rightarrow$ increased conductivity

- * Acceptor : p-type semiconductor
e.g., Boron (3A) : accepts 1 extra electron



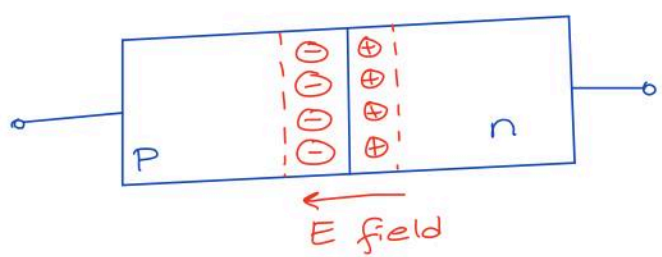
majority of carriers are holes
more doping $\Rightarrow$ increased conductivity

Drift and Diffusion Currents

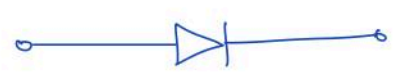
Drift : movement due to applied E field
(the type of current in standard conductors)

Diffusion : flow caused by concentration gradient of carriers

p-n Junction Silicon Diode



- * holes diffuse to n-region
- * electrons diffuse to p-region
- * E field created due to new charge distribution
 $\Rightarrow$ barrier potential for the carriers (V_0 threshold voltage)

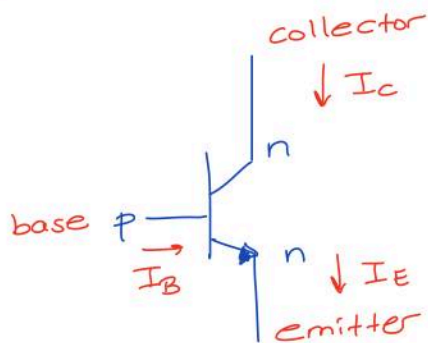


- * Reverse bias : further increases E field
 $\Rightarrow$ close to zero current
- * Forward bias : reduces E field
 $\Rightarrow$ diffusion current can be increased exponentially

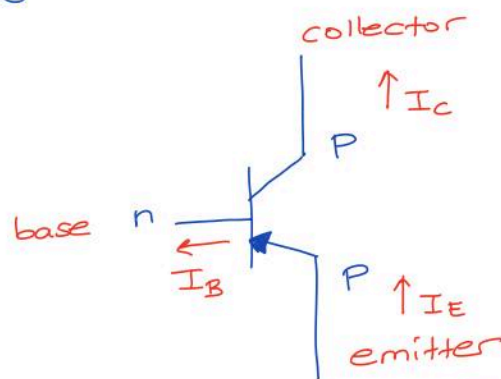
Bipolar Junction Transistor (BJT)

(10)

- * Used for modulation and RF amplification in TRC-II.
- * two p-n junctions connected back-to-back, sharing either p-region or n-region.

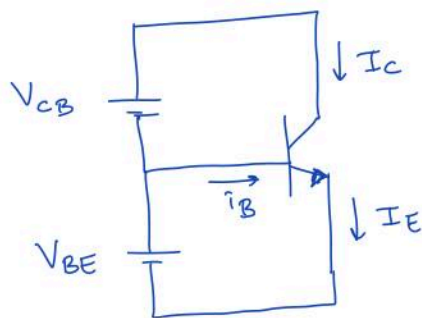
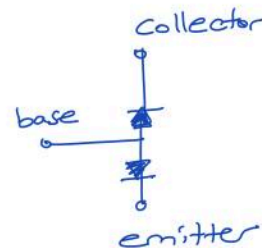
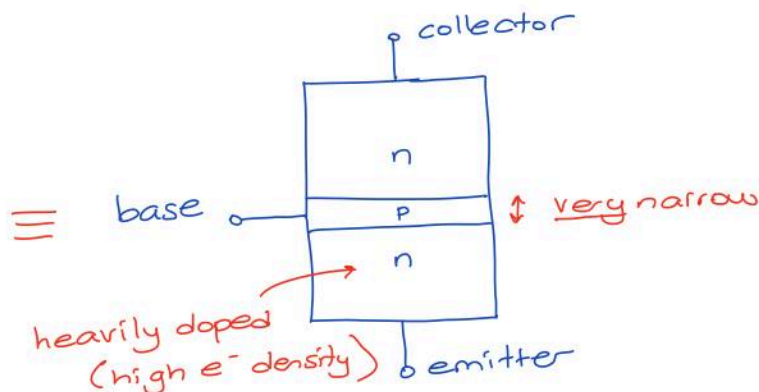
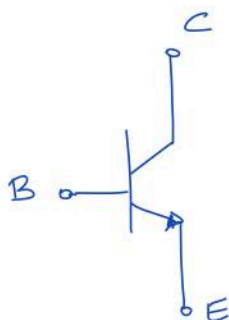


NPN type BJT
(majority of carriers are electrons)

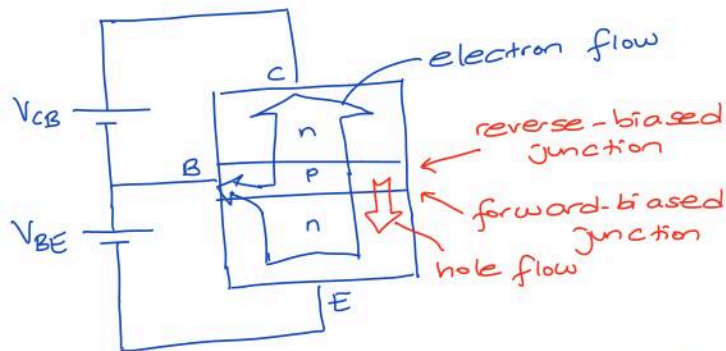


PNP type BJT
(majority of carriers are holes)

NPN BJT:



≡

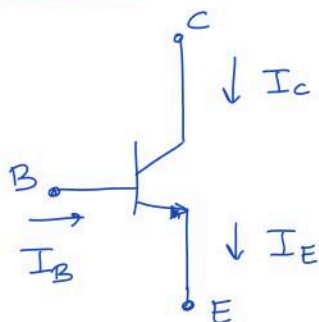


- * Electrons sent from emitter diffuse through the base and are collected at the collector.
- * A small fraction of electrons are captured in base.
- * There is also hole current from base to emitter.

* Overall, BJT is a current amplifier:
Base current is amplified at the collector.

(11)

NPN BJT:



current gain of BJT

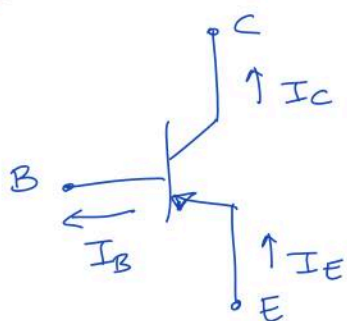
$$I_C = \beta \cdot I_B$$

$$I_E = I_B + I_C \quad (\text{KCL})$$

$$= I_B + \beta \cdot I_B$$

$$\text{So, } I_E = (\beta + 1) I_B$$

PNP BJT:



$$I_C = \beta I_B$$

$$I_E = (\beta + 1) I_B$$

States of a BJT:

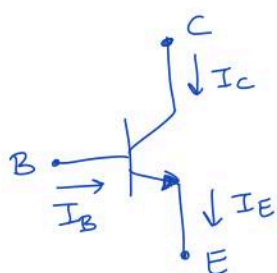
2 p-n junctions back-to-back
2 states for each $\Rightarrow$ 4 states overall

State	BE junction	BC junction
Cutoff (OFF)	reverse	reverse
Active (ACT)	forward	reverse
Reverse-active (REVACT)	reverse	forward
Saturation (SAT)	forward	forward

$\Leftarrow$ preferred state

$\Leftarrow$ not preferred (small β) should be avoided

* For the 3 common states:



State	V_{BE}	V_{CE}	I_B	I_C
OFF	$< V_0$	> 0	0	0
ACT	V_0	$> V_{CEsat}$	I_B	βI_B
SAT	V_0	V_{CEsat}	I_B	$< \beta I_B$

V_0 : turn-on voltage of BE diode. $V_0 \sim 0.6 - 0.8 \text{ V}$

V_{CEsat} : CE saturation voltage. $V_{CEsat} \sim 0.1 - 0.3 \text{ V}$

DC Analysis of BJT circuits

(12)

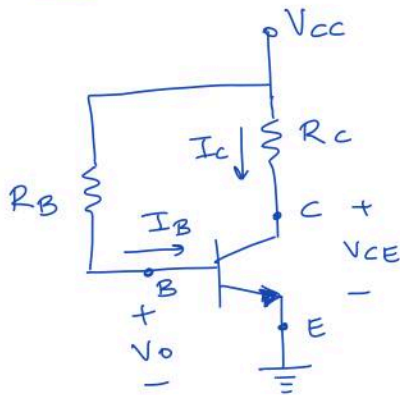
* BJT should be biased properly to ensure that it operates in ACT state.

* Similar to diode analysis:

- Assume a state for BJT
- Solve the corresponding linear circuit
- Check if the conditions for the assumed state are satisfied
- If not, try another state.

Simple Base Bias: To be in ACT state

BE : forward biased
BC : reverse biased



$$V_{BE} = V_0$$

$$I_B = \frac{V_{CC} - V_0}{R_B} > 0 \Rightarrow V_{CC} > V_0 \text{ to be in ACT state}$$

$$I_C = \beta \cdot I_B$$

$$V_{CE} = V_{CC} - I_C \cdot R_C > V_{CEsat} \text{ to be in ACT state}$$

* If $V_{CE} > V_{CEsat}$ not satisfied
 $\Rightarrow$ BC diode turns ON $\Rightarrow$ SAT state

Then,

$$I_C = \frac{V_{CC} - V_{CEsat}}{R_C} < \beta \cdot I_B$$

Example: $R_B = 470 \text{ k}\Omega$, $R_C = 1 \text{ k}\Omega$, $V_{CC} = 12 \text{ V}$
 $V_0 = 0.7 \text{ V}$, $V_{CEsat} = 0.2 \text{ V}$, $180 \leq \beta \leq 460$
 β can vary wildly from device to device.
We should ensure biasing circuit still works.

* For $\beta = 180$, assume ACT state

$$I_B = \frac{V_{CC} - V_0}{R_B} = \frac{12 - 0.7}{470 \text{ k}} = 24.04 \mu\text{A}$$

$$I_C = \beta \cdot I_B = 4.33 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C \cdot R_C = 12 - 4.33 \text{ mA} \times 1 \text{ k} = 7.67 \text{ V}$$

Check: $V_{CE} = 7.67 \text{ V} > V_{CEsat} = 0.2 \text{ V}$ ✓ ACT state

* For $\beta = 460$, assume ACT state

$$I_B = 24.04 \mu A \text{ (same)}$$

$$I_C = \beta \cdot I_B = 11.06 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C \cdot R_C = 12 - 11.06 \text{ mA} \times 1 \text{ k} = 0.94 \text{ V}$$

$$\text{Check: } V_{CE} = 0.94 \text{ V} > V_{CEsat} = 0.2 \text{ V} \quad \checkmark \text{ ACT state}$$

Example: Same circuit, increase R_C to $R_C = 1.2 \text{ k}\Omega$.

$\Rightarrow V_{CE}$ will get smaller

$\Rightarrow$ BJT is more likely to go out of ACT and into SAT state.

* For $\beta = 180$, assume ACT state

$$I_B = 24.04 \mu A \text{ (same)}$$

$$I_C = \beta \cdot I_B = 4.33 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C \cdot R_C = 6.8 \text{ V} > 0.2 \text{ V} \quad \checkmark \text{ ACT state}$$

* For $\beta = 460$, assume ACT state

$$I_B = 24.04 \mu A \text{ (same)}$$

$$I_C = \beta \cdot I_B = 11.06 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C \cdot R_C = -1.3 \text{ V} < 0.2 \text{ V} \quad \times \text{ not in ACT state}$$

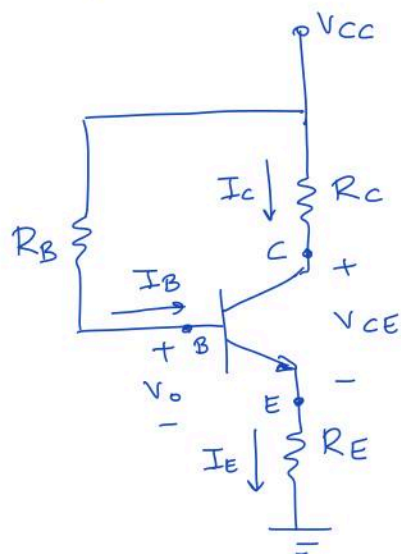
So, BJT in SAT state: $V_{CE} = V_{CEsat} = 0.2 \text{ V}$

$$I_C = \frac{V_{CC} - V_{CEsat}}{R_C} = 9.8 \text{ mA} < \beta I_B$$

we could not get the current gain we desired.

Base Bias with Emitter Resistor

- * β varies from device to device in a large range.
- * We want V_{CE} to not vary too much depending on β .
- * We need better circuits to guarantee that BJT operates in ACT state.



$$V_{CC} = I_B R_B + V_{BE} + I_E R_E \quad (\text{KVL})$$

$$I_E = (\beta + 1) I_B$$

So,

$$V_{CC} = I_B R_B + V_{BE} + (\beta + 1) I_B R_E$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E}$$

$$V_{CE} = V_{CC} - I_C R_C - I_E R_E$$

Check: $V_{CE} > V_{CEsat}$ for ACT state

- * If $V_{CE} > V_{CEsat}$ not satisfied $\Rightarrow$ SAT state

$$V_{CE} = V_{CEsat}$$

$$I_B + I_C = I_E \quad (\text{KCL})$$

$$\frac{V_{CC} - (V_{BE} + V_E)}{R_B} + \frac{V_{CC} - (V_{CEsat} + V_E)}{R_C} = \frac{V_E}{R_E}$$

Find V_E . Then,
$$I_C = \frac{V_{CC} - V_{CEsat} - V_E}{R_C}$$

Example: $R_B = 560 \text{ k}\Omega$, $R_C = 1 \text{ k}\Omega$, $R_E = 470 \Omega$
 $V_{CC} = 12 \text{ V}$, $V_{BE} = 0.7 \text{ V}$, $V_{CEsat} = 0.2 \text{ V}$
 $180 \leq \beta \leq 460$

* For $\beta = 180$: $I_B = 17 \mu\text{A}$, $V_{CE} = 7.4 \text{ V} > V_{CEsat}$ ✓

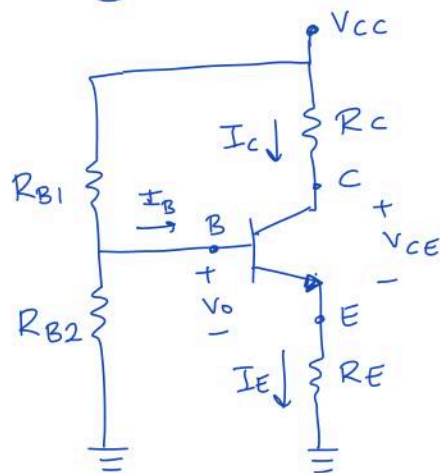
* For $\beta = 460$: $I_B = 14 \mu\text{A}$, $V_{CE} = 2.1 \text{ V} > V_{CEsat}$ ✓

Smaller variation in V_{CE} for different β .

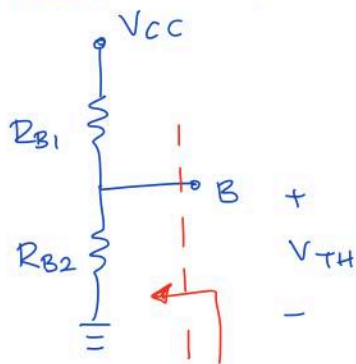
Conventional Bias Circuit

(15)

* A better biasing circuit:



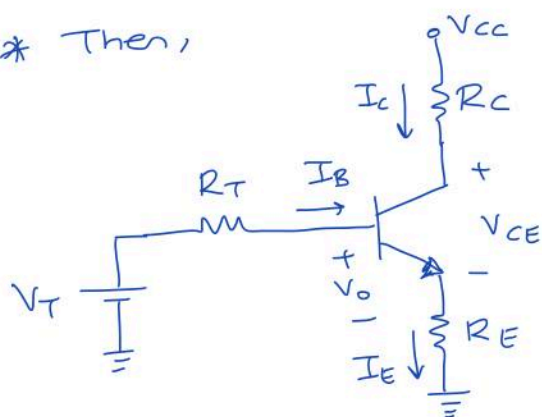
* Thevenin equivalent of the base side:



$$V_{TH} = V_{CC} \cdot \frac{R_{B2}}{R_{B1} + R_{B2}}$$

$$R_{eq} = R_T = R_{B1} \parallel R_{B2}$$

* Then,



$$V_T = I_B \cdot R_T + V_0 + I_E \cdot R_E$$

$$I_E = (\beta + 1) I_B$$

So,

$$I_B = \frac{V_T - V_0}{R_T + (\beta + 1) R_E}$$

Example: $R_{B1} = 18 \text{ k}\Omega$, $R_{B2} = 4.7 \text{ k}\Omega$, $R_C = 1 \text{ k}\Omega$, $R_E = 470 \Omega$
 $V_{CC} = 12 \text{ V}$, $V_0 = 0.7 \text{ V}$, $V_{CEsat} = 0.2 \text{ V}$

$$180 \leq \beta \leq 460$$

$$V_T = 12 \cdot \frac{4.7}{18 + 4.7} = 2.5 \text{ V}, \quad R_T = 18 \text{ k} \parallel 4.7 \text{ k} = 3.7 \text{ k}\Omega$$

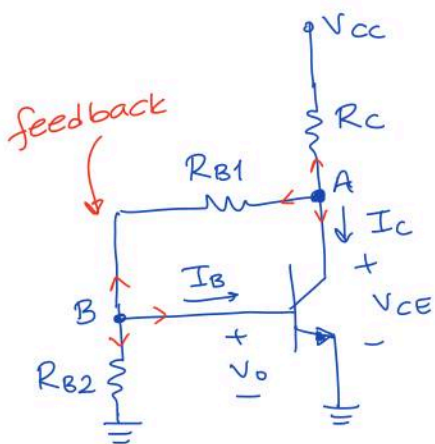
* For $\beta = 180$: $I_B = 20 \mu\text{A}$, $V_{CE} = 6.6 \text{ V} > V_{CEsat}$ ✓

* For $\beta = 460$: $I_B = 8.1 \mu\text{A}$, $V_{CE} = 6.5 \text{ V} > V_{CEsat}$ ✓

much smaller variation in V_{CE} for different β .

Bias Circuit with Collector Feedback

Assume BJT in ACT State.



$$V_A = V_{CE}, \quad V_B = V_o$$

$$\text{KCL at A: } \frac{V_{CE} - V_{CC}}{R_C} + I_C + \frac{V_{CE} - V_o}{R_{B1}} = 0 \quad (1)$$

$$\text{KCL at B: } \frac{V_o}{R_{B2}} + \underbrace{\frac{I_C}{\beta}}_{I_B} + \frac{V_o - V_{CE}}{R_{B1}} = 0 \quad (2)$$

$$\text{From Eqn. (2): } I_C = -\beta V_o \left(\frac{1}{R_{B1}} + \frac{1}{R_{B2}} \right) + \beta \frac{V_{CE}}{R_{B1}}$$

Substitute into Eqn. (1):

$$\frac{V_{CE} - V_{CC}}{R_C} - \beta V_o \left(\frac{1}{R_{B1}} + \frac{1}{R_{B2}} \right) + \beta \frac{V_{CE}}{R_{B1}} + \frac{V_{CE} - V_o}{R_{B1}} = 0$$

$$V_{CE} \left(\frac{1}{R_C} + \frac{\beta+1}{R_{B1}} \right) = \frac{V_{CC}}{R_C} + V_o \left(\frac{\beta+1}{R_{B1}} + \frac{\beta}{R_{B2}} \right)$$

Here, $V_{CE} > V_o > V_{CEsat}$

, if R_{B2} is not too small (makes I_C negative)

So, BJT is in ACT state.

* Ignoring small terms:

$$V_{CE} \left(\cancel{\frac{1}{R_C}} + \frac{\beta+1}{R_{B1}} \right) = \frac{V_{CC}}{R_C} + V_o \left(\frac{\beta+1}{R_{B1}} + \frac{\beta}{R_{B2}} \right)$$

ignore

$$\beta \frac{V_{CE}}{R_{B1}} = \frac{V_{CC}}{R_C} + \beta V_o \left(\frac{1}{R_{B1}} + \frac{1}{R_{B2}} \right)$$

$$V_{CE} = \underbrace{\frac{R_{B1}}{\beta \cdot R_C} \cdot V_{CC}}_{\text{too small}} + V_o \left(1 + \frac{R_{B1}}{R_{B2}} \right)$$

$$V_{CE} \approx \frac{R_{B1} + R_{B2}}{R_{B2}} \cdot V_o, \quad \text{if } R_C > \frac{R_{B1}}{\beta}$$

* Equivalent to ignoring the base current I_B .

* We should also check:

$V_{CE} < V_{CC}$: collector voltage cannot be larger than V_{CC} .

Example: $R_{B1} = 33 \text{ k}\Omega$, $R_{B2} = 4.7 \text{ k}\Omega$, $R_C = 1 \text{ k}\Omega$,
 $V_{CC} = 12 \text{ V}$, $V_o = 0.7 \text{ V}$, $V_{CEsat} = 0.2 \text{ V}$,
 $180 \leq \beta \leq 460$.

* For $\beta = 180$: $V_{CE} = 6.6 \text{ V}$, $I_C = 5.3 \text{ mA}$, $I_B = 29 \mu\text{A}$

* For $\beta = 460$: $V_{CE} = 6.0 \text{ V}$, $I_C = 5.8 \text{ mA}$, $I_B = 13 \mu\text{A}$

Small variation in V_{CE} , despite the large variation in β .

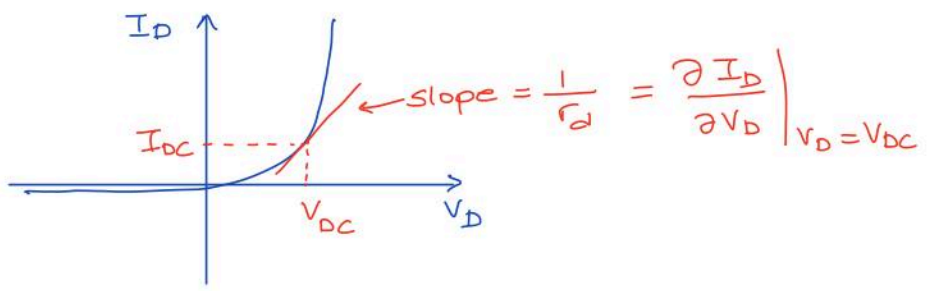
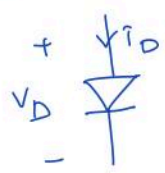
Small-Signal BJT model:

* If a small AC signal is added on top of a DC bias, we calculate the BJT response by linearizing it around the DC operating point.

* Equivalent to first-order Taylor series expansion

* Small-signal model is valid when the BJT always stays in the same state (due to proper DC biasing), and the applied AC signal is small.

Example: Diode



$$V_D = V_{DC} + \underbrace{(I_D - I_{DC})}_{i_D : \text{AC current}} \cdot r_d \quad \left(\text{First-order Taylor series expansion} \right)$$

$$f(x) = f(a) + f'(a)(x-a)$$

$$I_D = I_{DC} + \underbrace{i_D}_{\text{AC current through diode}}$$

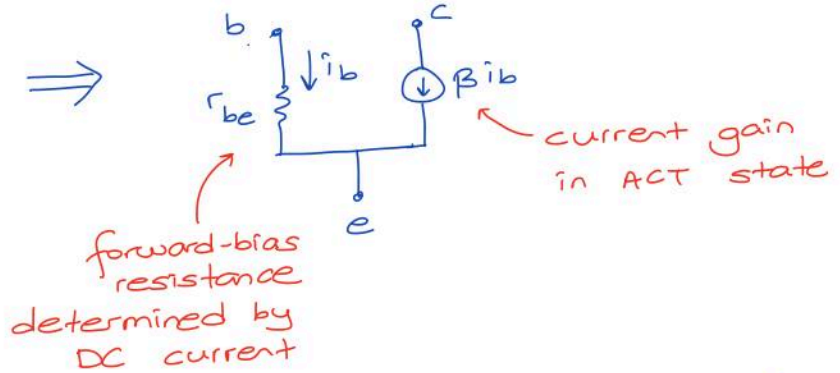
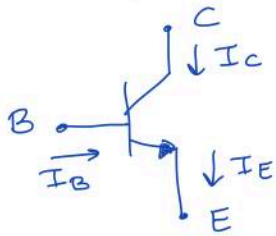
So,

$$V_D = \underbrace{V_{DC}}_{\text{DC response}} + \underbrace{i_D \cdot r_d}_{\text{AC response}} \quad \left(\text{superposition of DC + AC responses} \right)$$

So, Small-signal model for forward-bias (ON) state: $\left. \begin{array}{c} \vdots \\ r_d \end{array} \right\} : \text{resistor}$

Small-signal model for a BJT in ACT state:

- Useful to determine AC gain of BJT



* more complicated models exist (see EEE 313), but this model is sufficient for our purposes.

$$I_B = I_S \cdot e^{\frac{V_{BE}}{V_T}}$$

$$\frac{1}{r_{be}} = \frac{\partial I_B}{\partial V_{BE}} = \frac{1}{V_T} \cdot \underbrace{I_S e^{\frac{V_{BE}}{V_T}}}_{I_B} = \frac{I_B}{V_T}$$

$$V_T = \frac{\underbrace{k}_{\text{Boltzmann constant}} T}{\underbrace{Q}_{\text{electron charge}}} = \frac{1.38 \times 10^{-23} \times 300}{1.6 \times 10^{-19}} = 0.0259 \text{ V} = 25.9 \text{ mV}$$

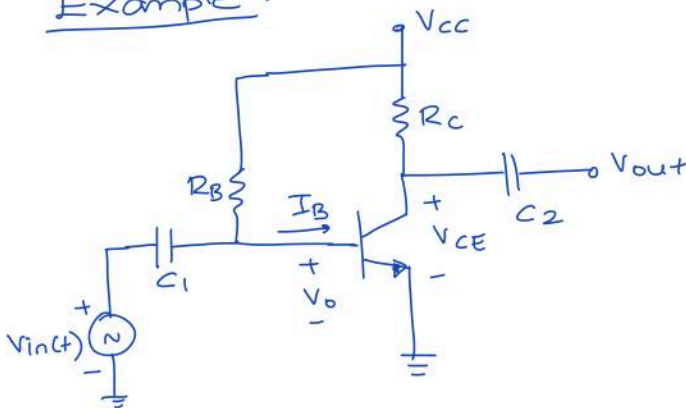
(thermal voltage)

So,

$$r_{be} (\Omega) = \frac{0.0259 \text{ V}}{I_B \text{ (A)}} = \frac{25.9 \text{ mV}}{I_B \text{ (mA)}}$$

↑
DC current through BE junction

Example:



$$R_B = 470 \text{ k}\Omega, R_C = 1 \text{ k}\Omega$$

$$V_{CC} = 12 \text{ V}, V_o = 0.7 \text{ V},$$

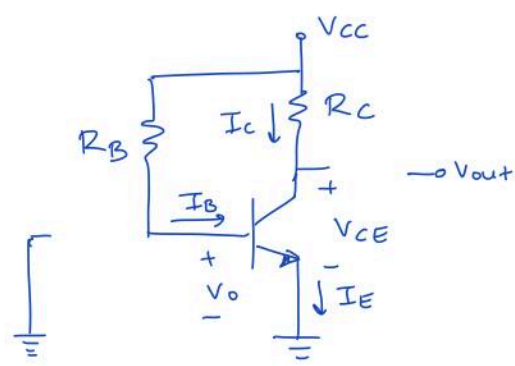
$$180 \leq \beta \leq 460$$

C_1, C_2 : DC block capacitors (large capacitors)

C_1 : prevents the bias circuit to be affected by V_{in}

C_2 : prevents the bias circuit to be affected by load.

DC analysis: kill $V_{in}(t)$, C_1 & C_2 act like open circuit



We analyzed this circuit already.
(simple base bias circuit)
Remains in ACT state for all β values.

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12 - 0.7}{470k} = 0.024mA$$

$V_{out,DC} = 0$ (no DC at output, thanks to C_2)

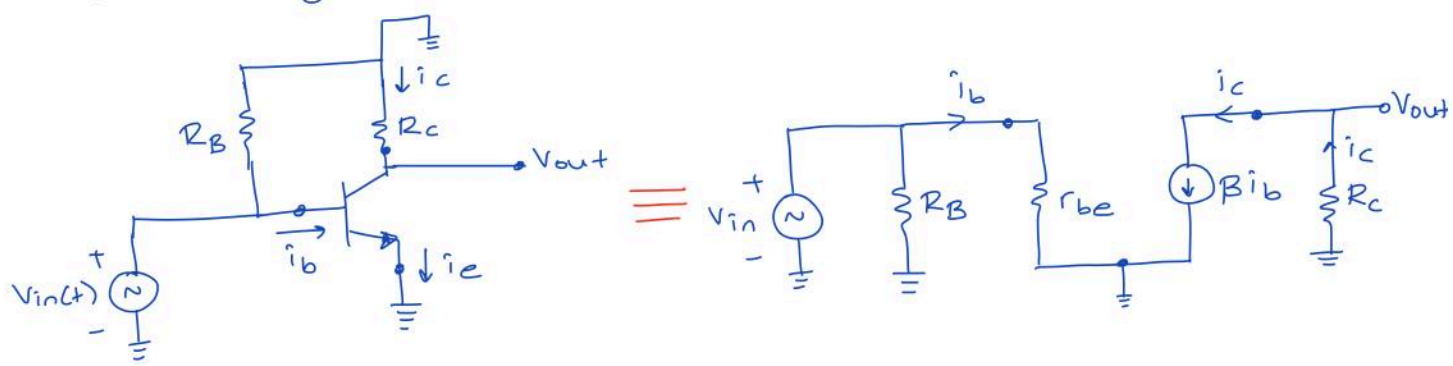
AC analysis: compute small-signal resistance r_{be}

$$r_{be} = \frac{25.9mV}{0.024mA} = 1079\Omega = 1.079k\Omega$$

C_1 & C_2 : assume they are sufficiently large at the AC input frequency

$\frac{1}{j\omega C}$ } very small $\Rightarrow C_1$ & C_2 behave like short circuit
large enough

Small signal AC circuit: kill V_{CC} , use small-signal BJT model



$$i_b = \frac{V_{in}}{r_{be}}$$

$$V_{out} = -i_c \cdot R_C = -\beta i_b \cdot R_C = -\beta \frac{V_{in}}{r_{be}} \cdot R_C$$

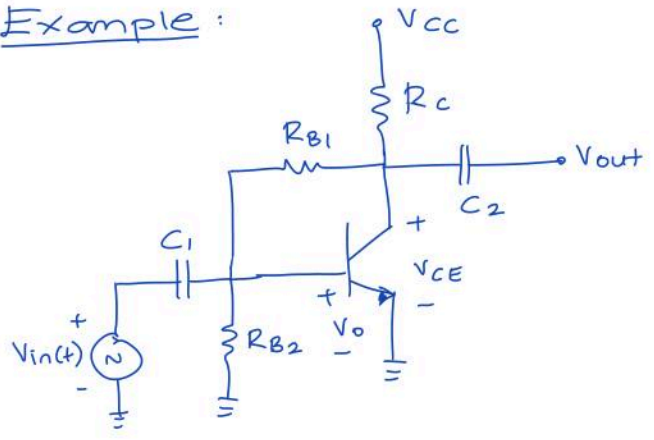
So, voltage gain: $A_V = \frac{V_{out}}{V_{in}} = -\beta \cdot \frac{R_C}{r_{be}}$

$$A_V = \begin{cases} -180 \cdot \frac{1k\Omega}{1.079k\Omega} = -167 & \text{for } \beta = 180 \\ -460 \cdot \frac{1k\Omega}{1.079k\Omega} = -426 & \text{for } \beta = 460 \end{cases}$$

indicates sign reversal (i.e., 180° phase change from input to output)

* β varies from device to device, so voltage gain will also vary. Hence, this is not an ideal circuit.

Example:



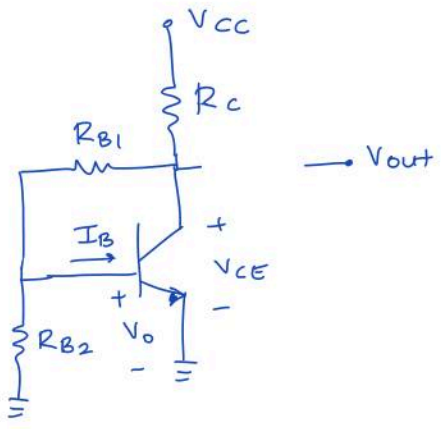
$$R_{B1} = 33k\Omega, R_{B2} = 4.7k\Omega, R_C = 1k\Omega$$

$$V_{CC} = 12V, V_{BE} = 0.7V$$

$$180 \leq \beta \leq 460$$

C_1, C_2 : DC block capacitors, sufficiently large to act like short circuit at AC input frequency.

DC analysis: kill $V_{in}(t)$. C_1 & C_2 act like open circuit



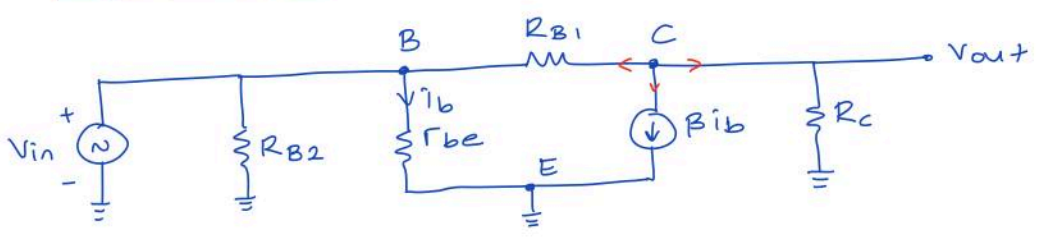
We analyzed this circuit already.
(bias circuit with collector feedback)
Remains in ACT state for all β values.

We previously found:

$$I_B = 29\mu A = 0.029mA \text{ for } \beta = 180$$

$$I_B = 13\mu A = 0.013mA \text{ for } \beta = 460$$

AC analysis: kill V_{CC} , C_1 & C_2 act like short circuit



$$r_{be} = \frac{25.9mV}{0.029mA} = 893\Omega = 0.89k\Omega \text{ for } \beta = 180$$

$$r_{be} = \frac{25.9mV}{0.013mA} = 1992\Omega = 1.99k\Omega \text{ for } \beta = 460$$

KCL at C: $\frac{V_{out}}{R_C} + \frac{V_{out} - V_{in}}{R_{B1}} + \beta i_b = 0$

Also: $i_b = \frac{V_{in}}{r_{be}}$

Insert: $\frac{V_{out}}{R_C} + \frac{V_{out} - V_{in}}{R_{B1}} + \beta \cdot \frac{V_{in}}{r_{be}} = 0$

$$V_{out} \left(\frac{1}{R_C} + \frac{1}{R_{B1}} \right) + V_{in} \left(\frac{\beta}{r_{be}} - \frac{1}{R_{B1}} \right) = 0$$

So, voltage gain is:

$$A_v = \frac{v_{out}}{v_{in}} = - \frac{\beta/r_{be} - \overbrace{1/R_B}^{\text{ignore}}}{1/R_C + 1/R_B} \approx - \frac{\beta}{r_{be}} \cdot \frac{1}{1/R_C + 1/R_B}$$

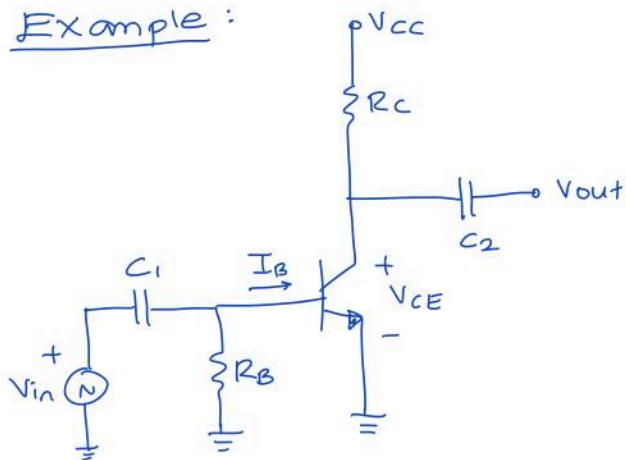
$$\approx - \beta \cdot \frac{R_C \parallel R_B}{r_{be}}$$

$$A_v = \begin{cases} -180 \cdot \frac{33 \parallel 1}{0.89} = -196 & , \text{ for } \beta = 180 \\ -460 \cdot \frac{33 \parallel 1}{1.99} = -224 & , \text{ for } \beta = 460 \end{cases}$$

The variation in gain is quite small, despite large variation in β .

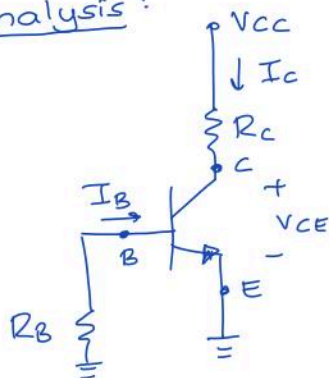
Note: If BJT is in SAT or OFF state, no AC voltage gain should be expected. So, a good biasing circuit is essential to get a small-signal voltage gain from a BJT.

Example:



$$R_B = 10k\Omega, R_C = 1k\Omega, \\ V_C = 12V, V_D = 0.7V, V_{CEsat} = 0.2V$$

DC analysis:



Bias circuit is faulty.

No positive bias is supplied to the base.

BE diode is OFF.

$$I_B = 0, V_B = 0 < V_D \quad (\text{confirms } V_{BE} \text{ OFF})$$

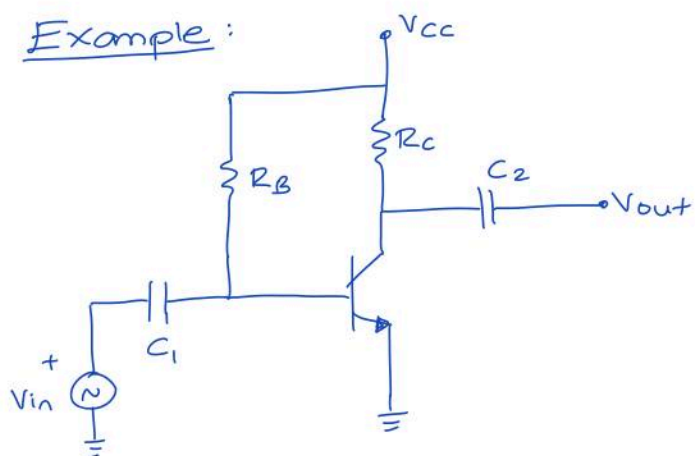
V_{BC} reverse biased.

$$I_C = 0, V_{CE} = V_{CC} > 0$$

So, BJT is in OFF state.

Small-signal gain is zero.

Example:

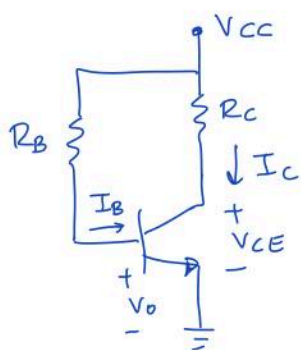


$$R_B = 10\text{ k}\Omega, R_C = 1\text{ k}\Omega,$$

$$V_{CC} = 12\text{ V}, V_{BE} = 0.7\text{ V}, V_{CE\text{sat}} = 0.2\text{ V}$$

$$\beta = 150$$

DC analysis: Assume that BJT is in ACT state:



$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = 1.13\text{ mA}$$

$$I_C = \beta I_B = 150 \times 1.13\text{ mA} = 169.5\text{ mA}$$

very large!

$$V_{CE} = V_{CC} - I_C R_C = 12 - 169.5\text{ mA} \times 1\text{ k}\Omega$$

$$= -157.5\text{ V} < V_{\text{sat}} \quad \times \text{ cannot happen}$$

So, BJT is in SAT state.

Small-signal gain is zero.